Mapping Natural-Disaster Damage from Satellite Imagery

Cross-disaster transfer of a pretrained change-detection CNN on xBD

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Abstract. We assess per-building damage from paired pre- and post-disaster satellite imagery on the xBD dataset, and measure whether a model trained on one family of disasters transfers to a family it has never seen. A siamese CNN built on an ImageNet-pretrained ResNet-18 is trained on 14 non-fire events and evaluated both in-distribution and zero-shot on five wildfire events held out entirely. In-distribution the model reaches **0.775 accuracy / 0.637 macro-F1**; on unseen wildfire it drops to **0.636 / 0.396**, a 37.9 % relative loss in macro-F1. The per-class breakdown shows this headline number is misleading. destroyed scores *higher* out of domain than in it (F1 0.754 vs. 0.631) and no-damage precision reaches 0.977; the collapse is confined to the two intermediate classes, which make up 19.1 % of training patches but only 1.7 % of wildfire patches. Together with a strongly asymmetric error direction (85.6 % of out-of-domain errors over-estimate severity) and a threshold-free ROC-AUC that holds at 0.797, the evidence indicates that the learned features transfer while the decision calibration does not. We also report a negative result on the team's data-quality question: xBD carries no occlusion label, and a four-statistic diagnostic finds the worst-occluded tiles are cloud over tsunami and volcano scenes rather than smoke over wildfire.

Scope of this draft. This report covers Part 1 (data collection, cleaning and preliminary analysis) and the *pretrained* half of Part 2. The from-scratch CNN baseline and Part 3 (impact and recovery analysis) are in progress and are not reported here. All numbers below come from a single reproducible run; nothing is projected or estimated.

1 Introduction

Rapid damage maps after a natural disaster are useful only if they can be produced *before* labelled data for that disaster exists. That makes the operational question sharper than ordinary image classification: not "how accurate is the model" but "how accurate is it on a disaster type it has never seen".

We study this on xBD [1], which pairs pre- and post-event high-resolution satellite tiles with building footprints and a four-level damage scale. The task is framed as *change detection at building level*: given two co-registered crops of the same structure, one before and one after the event, predict no-damage, minor-damage, major-damage or destroyed.

The experiment deliberately avoids a random split. Five wildfire events are removed from training entirely and used as a zero-shot test set, while the official xView2 evaluation directories provide an in-distribution reference measured on the same trained model. The *gap* between those two numbers is the result; neither is informative alone.

2 Data

2.1 Source and access

We use the full xBD release: 11,034 scenes over 19 events, each scene contributing a 1024×1024 RGB pre-disaster tile, the matching post-disaster tile, and a GeoJSON label file. That is 22,068 images and **425,368** annotated building polygons on the post-disaster side. Imagery was obtained through a public mirror of the xView2 challenge release[1].

Two properties of the source matter and were verified rather than assumed:

- **The tiles are already co-registered.** Pixel (x, y) is the same ground location in the pre- and post-disaster image, which is what makes cropping the *same* bounding box from both valid. Without it the siamese design would be comparing different places.
- **The format is uniform.** All 800 files in a stratified quality-control sample are 1024×1024 RGB PNG with no read errors, so no resolution standardisation or band alignment is required.

Damage labels live only in the post-disaster GeoJSON, under `features.xy[].properties.subtype`. The same file also carries `features.lng_lat`, giving per-building WGS84 coordinates directly — which means the geo-referencing needed for Part 3 is already present and no separate geotransform file is required.

2.2 Split design

The split is scene-level and was produced by a dedicated indexing script rather than inside the training notebook, so that every model in the project trains on an identical partition. Assignment rules and resulting sizes are given in Table 1.

Two design points are worth defending. First, `test_id` is drawn from the official `test` and `hold` directories, which were never part of any training pool and, unlike `val`, were never used for tuning — this makes them an honest reference for the headline number. Second, assignment happens at *scene* level, which is what

[1] Kaggle dataset `qianlanzz/xbd-dataset`, layout `xbd/{hold,test,tier1,tier3}/{images,labels,masks}`.

Table 1: Split design and sizes. Only `train`/`val` depend on the random seed; both test sets are defined by the data itself.

Split	Rule	Scenes	Polygons
`train`	tier1+tier3, non-fire	3,135	225,506
`val`	tier1+tier3, non-fire	392	30,794
`test_id`	test+hold, non-fire	1,135	91,755
`test_ood`	all wildfire scenes	6,372	77,313
Total		11,034	425,368

guarantees that the pre- and post-disaster image of a location cannot separate and that no building can appear in two splits.

The five held-out events are `pinery-bushfire`, `portugal-wildfire`, `santa-rosa-wildfire`, `socal-fire` and `woolsey-fire`. An automated check confirms zero wildfire scenes in `train` or `val` and zero event overlap between the training pool and `test_ood`.

3 Exploratory analysis

3.1 Class imbalance is severe and split-dependent

no-damage accounts for 72.0 % of training polygons and the rarest class is 11.6× smaller (Fig. 1). More importantly the composition is not stable across splits: `test_ood` is 80.2 / 0.9 / 0.9 / 14.7 % against 72.0 / 9.7 / 8.4 / 6.2 % in `train`.

This has direct methodological consequences. Plain accuracy is close to useless — always predicting no-damage scores 0.72 on the training distribution while producing a damage map that never flags damage. We therefore report macro-F1 and per-class recall as the decision metrics, use inverse-frequency class weights computed on `train` only (0.33/2.38/2.91/4.92), and key checkpointing and early stopping on validation macro-F1 rather than accuracy.

Class weighting was preferred to resampling: oversampling would repeat the same few destroyed buildings many times per epoch and invite memorisation, while undersampling would discard most of the majority signal. Weighting keeps every sample once per epoch and alters only its gradient contribution.

3.2 Wildfire damage is bimodal

Averaged over events, the non-fire domain is 70.2 / 9.9 / 7.2 / 6.5 % across the four classes while the wildfire domain is 78.8 / **1.2** / **1.1** / **14.8** %. Fire either leaves a building intact or destroys it; intermediate states barely occur. Figure 2 shows this per event and also how much the non-fire events differ among themselves — `mexico-earthquake` is 99 % undamaged, `hurricane-matthew` is 51 % minor-damage.

This is the project's central hypothesis made measurable. A model that implicitly learns the hurricane prior will be miscalibrated on fire *even if its visual features transfer perfectly.* It also tells us what to look for in the out-of-domain confusion matrix: a pure prior shift produces errors with a consistent direction, whereas failed features produce diffuse errors.

3.3 Scene occupancy

Empty scenes are common and strongly event-dependent (Fig. 3): 9.8 % of training scenes contain no buildings at all, against **54.1 %** of wildfire scenes, whose median building count is zero. The tier3 fire events cover large stretches of forest and scrubland.

An empty scene yields no samples but still costs a full 1024 × 1024 decode. Sampling wildfire scenes uniformly would therefore have spent most of the extraction budget on empty forest. We instead sample only scenes with at least five usable buildings. This is a *sampling* decision, not a cleaning one: nothing is deleted, and the identical rule is applied to all four splits so it cannot tilt the comparison.

3.4 Footprint geometry sets the patch size

Buildings are small in absolute pixels: median longest bounding-box side 35 px, 90th percentile 64 px, with a thin tail to 349 px at the 99th (Fig. 4). A 64 × 64 patch therefore covers roughly 90 % of footprints at close to native resolution. Going larger mostly upsamples empty ground at quadratic cost; going smaller destroys the roof texture that separates minor from major damage.

We keep a 10 px margin around each footprint. Damage is frequently more visible *around* a building than on it — debris fields, scorch marks, standing water — but the margin is kept modest deliberately, for reasons that Section 7 makes concrete.

3.5 A confound in the acquisition geometry

Ground sample distance and off-nadir angle are similar across the two domains, but sun elevation is not: 64.4± 7.5 degrees for non-fire against 55.0 ± 15.9 for wildfire, with a fire mode near 35° that has no counterpart in training (Fig. 5). Low sun means long shadows and a materially different appearance for the same structure.

Honest caveat. The histograms are spiky because every scene within an event shares one acquisition. Statistically this is **19 points, not 11,034**, and the geometry is fully confounded with event identity — "wildfire" and "the captures that happen to be wildfire" cannot be separated on this data. Part of the transfer gap we report may be shadow length rather than fire damage, and we cannot bound how much.

3.6 What the classes look like

Figure 6 contrasts the two domains directly. In the non-fire domain the post-disaster image usually retains the

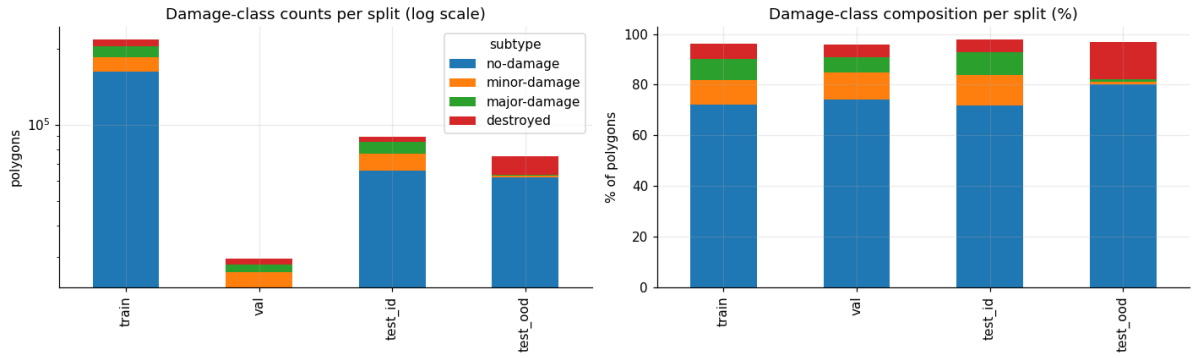


Figure 1: Damage-class distribution per split. Left: counts on a log scale. Right: composition in per cent. The `test_ood` bar is qualitatively different from the other three — the intermediate classes almost vanish and destroyed more than doubles.

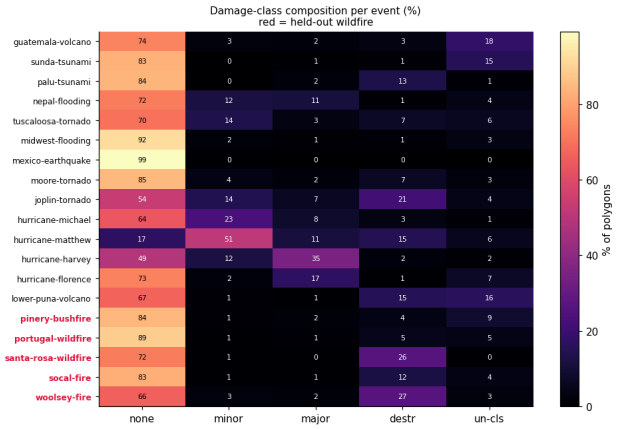


Figure 2: Damage-class composition per event. Held-out wildfire events in red. The fire rows share a signature that no non-fire event reproduces: near-zero minor/major with elevated destroyed.

structure with altered geometry — torn roofs, debris, standing water. Wildfire damage instead collapses a building to a uniform grey scar with the footprint faintly visible, while *undamaged* buildings often sit in surroundings that changed completely (scorched vegetation) with the roof untouched.

That last observation generates a falsifiable prediction, made before any model was trained: a change-detection model learns "the scene changed" ⇒ "the building is damaged", so under wildfire it should over-flag no-damage. Section 6 reports the outcome.

4 Data cleaning

Our operating principle is that nothing is removed silently. Every rule states its threshold, its cost, and its per-split removal rate, so that a rule which quietly favours one split can be caught. Table 2 is the full ledger.

The two rules that actually cost data are the un-classified subtype (3.29 %) and sub-8-pixel footprints (3.51 %). Both matter for fairness, so both were checked per split. The un-classified rate is 3.67 / 4.08 / 2.09 / 3.30 % across train / val / test_id / test_ood, and the tiny-footprint removal rate is 3.77 / 3.30 / 2.99 / 3.49 %. Neither is concentrated in `test_ood`, so neither flatters the transfer result.

Table 2: Cleaning ledger. Removal is reported against the scope each rule operates on.

Rule	Scope	Removed
Missing / unpaired files	11,034 scenes	0
Corrupted images	800 sampled images	0
Unreadable labels	11,034 scenes	0
Smoke / cloud occlusion	683 scenes	0 (flagged)
un-classified subtype	425,368 polygons	14,011
Degenerate polygons	425,368 polygons	0
Footprint side < 8 px	411,357 polygons	14,458
Near-duplicates	400 sampled scenes	0 (no leak)
Usable after cleaning		**396,899 (93.3 %)**

4.1 The occlusion question: a negative result

The team's open data-quality question was whether images fully obscured by smoke should be removed. **xBD carries no occlusion label** — the metadata block contains only capture geometry. The closest available proxy is the un-classified subtype, which annotators used when a building could not be assessed.

We ranked scenes by that proxy and tested it against four whole-tile image statistics chosen because smoke and thin cloud on 3-band optical imagery are bright, low-contrast, low-saturation and edge-poor. The measured Spearman correlations are given in Table 3 and Fig. 7.

Only edge density correlates meaningfully; brightness is weak and contrast is essentially zero. The composite ranking is therefore measuring "texture-poor", not "bright and flat" as hypothesised.

Table 3: Rank correlation between the annotator "cannot assess" rate and whole-tile image statistics ($n = 505$).

Statistic	Spearman ρ
Edge density	−0.366
Brightness	+0.116
Saturation	−0.095
Contrast	−0.035

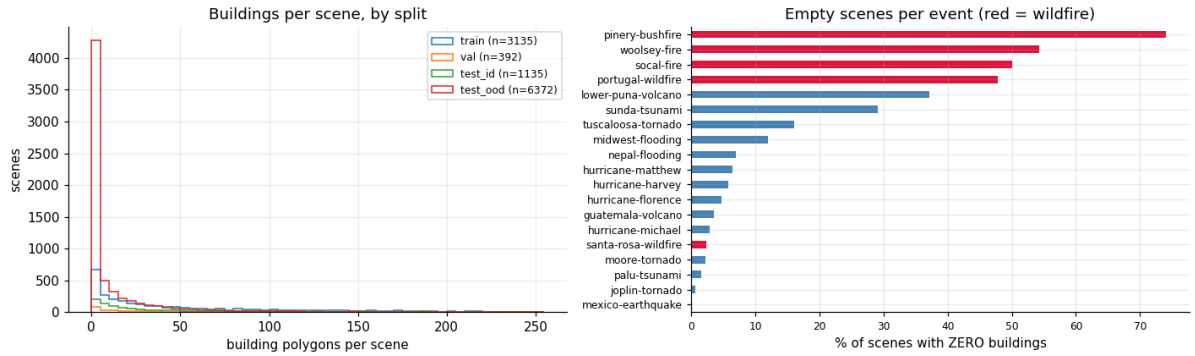


Figure 3: Left: distribution of building count per scene by split. Right: share of scenes with zero buildings, per event (wildfire in red). More than half of all wildfire scenes are empty.

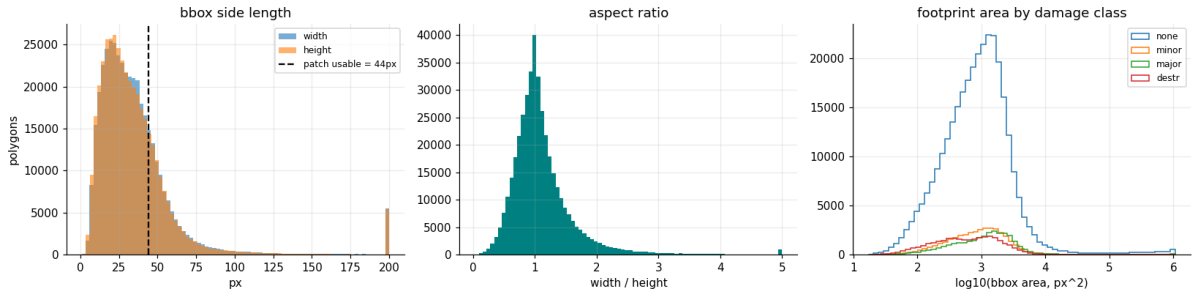


Figure 4: Building footprint geometry. Left: bounding-box side lengths against the usable patch area. Centre: aspect ratio. Right: footprint area by damage class. The dashed line marks the $64 - 2 \times 10$ px usable window.

Result, and why nothing was removed. The highest-ranked tiles are *cloud over tsunami and volcano scenes*, not smoke over wildfire (Fig. 8). At the 98th percentile the flagged set is 14 scenes: 11 in `train`, 3 in `val`, **0 in either test split**. We removed nothing, for four reasons. *(i)* There is no ground truth against which to validate a threshold, so any cut point would be tuned by eye and would introduce an unmeasurable bias. *(ii)* The affected volume is negligible and touches no reported metric. *(iii)* Occlusion is part of the deployment problem — a system running on real post-disaster imagery will meet cloud and smoke, and deleting it measures a scenario that does not exist. *(iv)* A safer patch-level filter already exists downstream, rejecting crops that are essentially pure white or black; in this run it rejected zero patches. Candidates are exported for manual review instead.

4.2 Leakage checks

A perceptual-hash duplicate test over a 400-scene sample returned 28 near-duplicate pairs, 19 of them crossing a split boundary. Investigated rather than waved through: every cross-split pair links *different events on different continents* — one `portugal-wildfire` tile matching `nepal-flooding`, `hurricane-harvey`, `sunda-tsunami`, `palu-tsunami` and `moore-tornado` simultaneously. A single tile cannot be the same location as five unrelated disasters. The cause is hash degeneracy: a 64-bit difference hash of a low-texture tile (open water, unbroken forest, solid cloud) collapses toward a constant, so flat tiles collide with each other regardless of content. We find no evidence of leakage, consistent with the scene-level split design.

Table 4: Extracted patch pairs per split and class. Note the support for minor and major in `test_ood`.

	none	minor	major	destr	**total**
`train`	55,284	7,661	6,265	3,706	72,916
`val`	15,701	2,571	1,164	1,010	20,446
`test_id`	20,500	2,840	2,512	1,421	27,273
`test_ood`	12,056	**116**	**138**	2,317	14,627
Total					**135,262**

5 Method

5.1 Patch construction

One sample is two $64 \times 64 \times 3$ `uint8` patches cropped from the *same* pixel bounding box in the pre- and post-disaster tile, plus one integer label. Because the tiles are already co-registered, no warping is required; preprocessing is crop, resize, scale to $[0, 1]$ and ImageNet normalisation.

Scenes were sampled under a compute budget, stratified by event and restricted to scenes with at least five usable buildings, yielding **135,262** patch pairs from 1,698 scenes (Table 4). Patches are written to memory-mapped arrays rather than individual image files, which keeps the working set on disk and the loader fast.

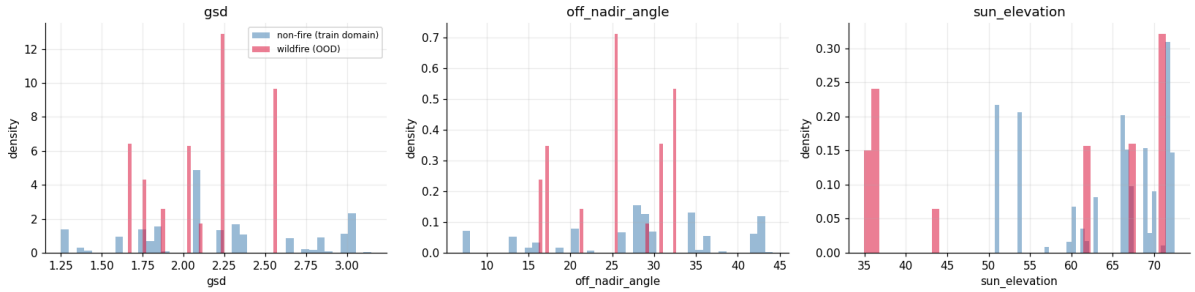
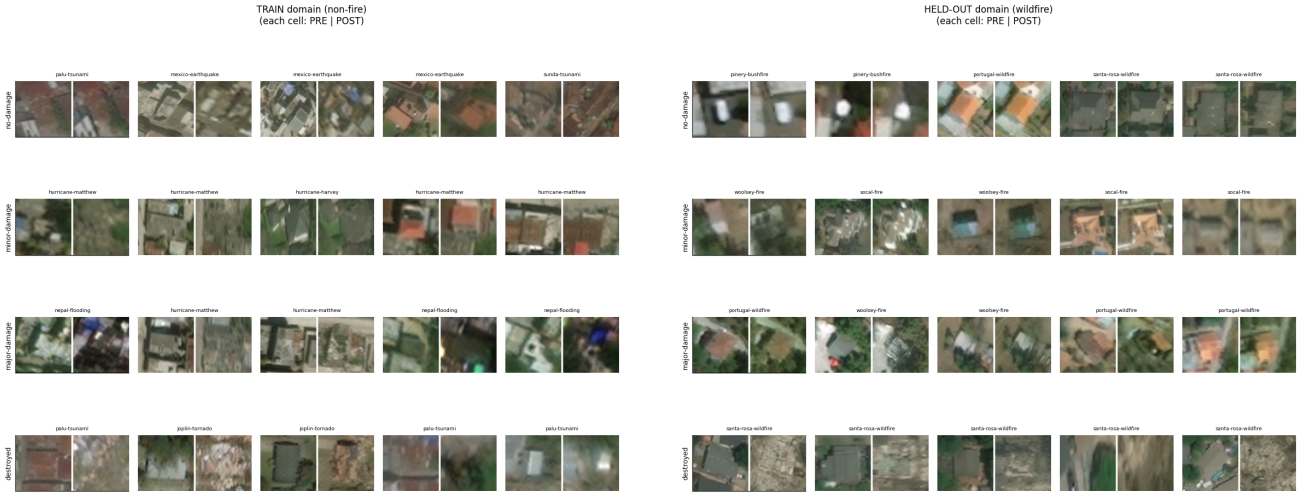


Figure 5: Capture metadata, non-fire (blue) against wildfire (red). GSD and off-nadir angle overlap; sun elevation does not.



(a) Training domain (non-fire). **(b)** Held-out domain (wildfire).

Figure 6: Representative building patches, one row per damage class. Each cell shows the pre-disaster crop on the left and the post-disaster crop on the right.

5.2 Architecture

The task is a comparison, not the classification of a single image. Using only the post-disaster patch discards the reference, while stacking pre and post into a six-channel input forces the first convolution to be relearned, throwing away the pretrained weights exactly where they transfer best. We therefore use a **siamese encoder with shared weights**, with a small head reading

$$\left[\, f_{\text{pre}},\ \ f_{\text{post}},\ \ |f_{\text{pre}} - f_{\text{post}}| \,\right].$$

The absolute difference is the explicit change signal; retaining the raw features alongside it lets the head use appearance as well. Shared weights halve the parameter count and force both branches into a common feature space, which is what makes the difference term meaningful.

The encoder is an ImageNet-pretrained ResNet-18 [3, 4]. At a 64×64 input, depth stops paying: architectures designed for 224 px inputs downsample away the little spatial resolution available, and a vision transformer would reduce a 64 px image to a handful of tokens without nearly enough data to compensate for the missing convolutional prior. ResNet-18 also keeps the comparison honest — the project's other model is a compact from-scratch CNN, and a much larger backbone would make any win attributable to model size rather than to pretraining.

5.3 Training

Training follows a linear-probe-then-fine-tune schedule [5]: phase 1 trains only the head on a frozen backbone for 4 epochs at 10^{-3}, so that the randomly initialised head does not push large gradients into good weights; phase 2 unfreezes `layer3` and `layer4` for 10 epochs at 10^{-4}. Early layers transfer essentially unchanged, while the late semantic layers are what must be repurposed. This ordering was chosen deliberately because it distorts pretrained features less, which matters when the headline result is out-of-domain performance.

Optimisation uses AdamW with cosine annealing, weighted cross-entropy, mixed precision, and checkpointing plus early stopping on validation macro-F1. Everything ran on a single Kaggle Tesla T4: patch extraction took 2.3 minutes and the 14 training epochs about 5.4 minutes.

The curves (Fig. 9) contain two observations worth recording. Unfreezing lifted validation macro-F1 from 0.477 to **0.580 in a single epoch** — the clearest evidence in this study that a frozen ImageNet backbone is not sufficient for satellite imagery and that fine-tuning is doing real work. Second, from fine-tune epoch 3 the validation *loss* rises ($0.588 \rightarrow 0.615$) while

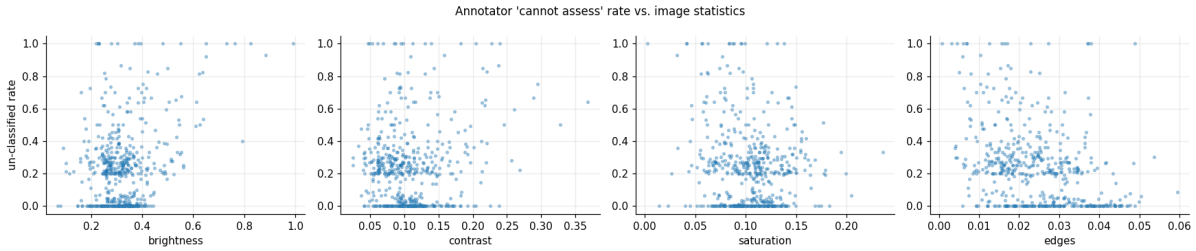


Figure 7: Annotator "cannot assess" rate against four whole-tile statistics. Only edge density (right) shows a usable relationship.

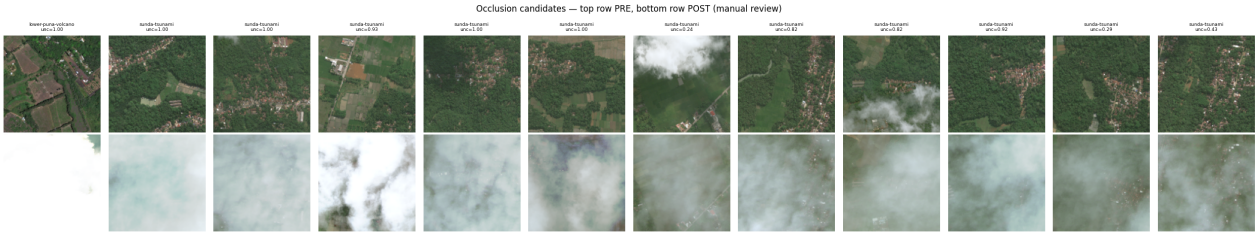


Figure 8: Highest-ranked occlusion candidates, pre-disaster on top and post-disaster below. These are cloud over `sunda-tsunami` and `lower-puna-volcano` scenes — not smoke over wildfire. The worst tile has brightness 0.99 at contrast 0.049.

Table 5: Headline results. One trained model, three evaluation sets.

	Accuracy	Macro-F1	ROC-AUC
`val` (seen)	0.793	0.640	0.916
`test_id` (seen)	0.775	0.637	0.917
`test_ood` (wildfire)	0.636	0.396	0.797
Absolute drop	−0.139	−0.241	−0.120
Relative drop	−17.9%	−37.9%	−13.0%

Table 6: Per-class F1 and out-of-domain precision. destroyed performs *better* out of domain than in it.

Class	F1 id	F1 ood	P ood	n ood
no-damage	0.877	0.765	0.977	12,056
minor-damage	0.502	**0.023**	**0.012**	116
major-damage	0.539	**0.042**	**0.022**	138
destroyed	0.631	**0.754**	0.801	2,317

macro-F1 continues to improve (0.623 → 0.640). Under a weighted loss on imbalanced data this is expected: the model grows over-confident faster than it grows wrong. It is also why loss would have been the wrong early-stopping criterion. Early stopping did not trigger; the best epoch was the 9th of 10.

6 Results

Table 5 gives the headline numbers. `val` and `test_id` agree to 0.003 macro-F1, which matters: it shows the in-distribution figure is not a lucky validation draw, so the gap is a genuine domain effect rather than noise.

6.1 Per-class behaviour

The aggregate drop is misleading, and the per-class view (Table 6, Fig. 10) shows why.

destroyed scores 0.754 on wildfire against 0.631 in domain, and no-damage precision reaches 0.977. A model that has never seen a fire recognises burnt-out buildings *better* than it recognises hurricane destruction. The collapse is confined entirely to the two intermediate classes, where precision falls to 0.012 and 0.022.

6.2 Error structure

Of all out-of-domain errors, **85.6 % over-estimate severity** and only 14.4 % under-estimate it (Fig. 11). Restricting attention to buildings that are truly undamaged, the model predicts no-damage for 62.8 %, minor for 19.4 %, major for 14.8 % and destroyed for 3.0 % — a **37.2 %** false-alarm rate. This confirms the prediction made from the exploratory analysis before training.

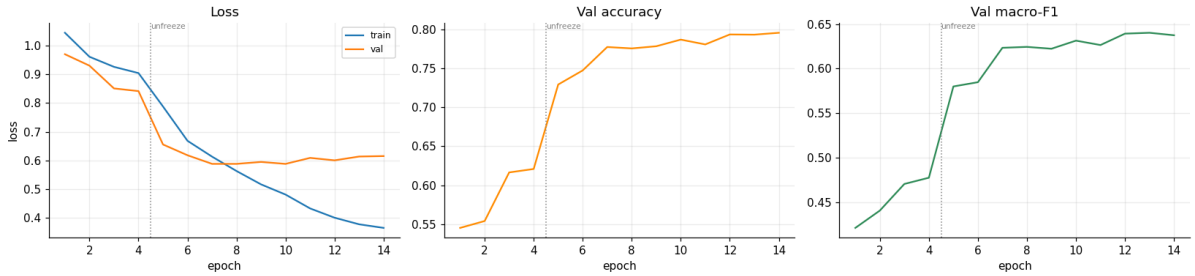


Figure 9: Training history. The dotted line marks the transition from frozen backbone to fine-tuning. Note that from fine-tune epoch 3 onward validation loss rises while validation macro-F1 keeps improving.

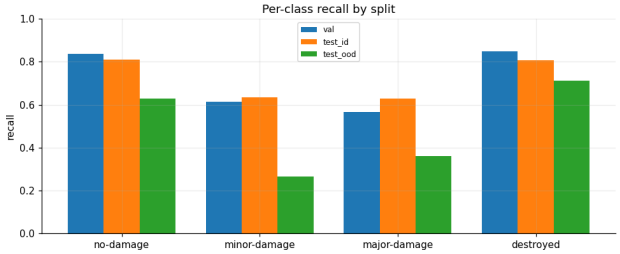


Figure 10: Per-class recall by split. Recall on the intermediate classes degrades out of domain but does not vanish; it is precision that collapses.

7 Discussion

The features transfer; the calibration does not. Four independent measurements support this reading of the 37.9 % macro-F1 drop. *(1)* destroyed scores higher out of domain than in it (0.754 vs. 0.631) and no-damage precision is 0.977 — the two classes that actually occur under wildfire are recognised well. *(2)* 85.6 % of errors move in one direction. A consistent bias is the signature of a shifted prior; failed representations would produce diffuse, symmetric confusion. *(3)* ROC-AUC, which does not depend on a decision threshold, holds at 0.797 against 0.917 in domain. The ranking survives; the boundary is what breaks. *(4)* The collapse is arithmetically forced. minor and major are 10.5 % and 8.6 % of training patches but 0.79 % and 0.94 % of wildfire patches. A classifier calibrated for a 10 % prior emits roughly 4,800 intermediate predictions against 254 true cases, capping precision near 0.02 regardless of feature quality.

The distinction is not academic. A representation failure would mean the approach cannot be deployed on a new disaster type at all. A calibration failure means the ranking is usable and the decision rule needs adjusting — a far more tractable problem, and one with an established remedy in prior correction [6], which rescales class posteriors by the ratio of target to source priors without retraining.

Operationally, a model like this deployed on an unlabelled new disaster type would reliably separate destroyed from intact structures while badly overreporting intermediate damage. For triage — "where should responders look first" — that failure mode is comparatively benign. For damage *quantification*, and certainly for aid allocation, it is not, and acting on predicted rather than verified damage carries a real risk of misdirecting resources.

The exploratory prediction about `BBOX_PAD` also lands here. Because context change misleads the model under wildfire, enlarging the crop margin would likely make transfer *worse*, not better — a testable claim we have not yet run.

8 Limitations

- **Thin out-of-domain support.** minor and major rest on 116 and 138 true examples in `test_ood`. These two classes drive the macro-F1 story, and their per-class figures carry wide uncertainty.
- **Annotator selection.** Every number is measured on buildings that human annotators were *able* to assess; the un-classified polygons are excluded. True zero-shot difficulty is therefore somewhat higher than reported.
- **Confounded acquisition geometry.** The sun-elevation shift is real but inseparable from event identity with only 19 events.
- **Partial duplicate check.** The near-duplicate test covers 400 of 11,034 scenes and relies on a perceptual hash that degenerates on low-texture tiles.
- **Selection protocol.** Model selection uses `val`, which is in-distribution. Longer fine-tuning may improve `test_id` while harming `test_ood`; selecting on the out-of-domain set would leak the test set, so this is the correct protocol but a genuine constraint.
- **Budgeted sampling.** Results are computed on a stratified sample of scenes rather than the full dataset.

9 Status and next steps

Completed: Part 1 in full, and the pretrained model of Part 2. Outstanding work, in priority order:

1. **From-scratch CNN baseline**, trained on the identical patch arrays so that the comparison isolates pretraining rather than architecture or data.
2. **Test the calibration hypothesis directly.** Apply prior correction [6] to the existing out-of-domain predictions and measure how much of the 37.9 % macro-F1 gap closes without retraining. If most of

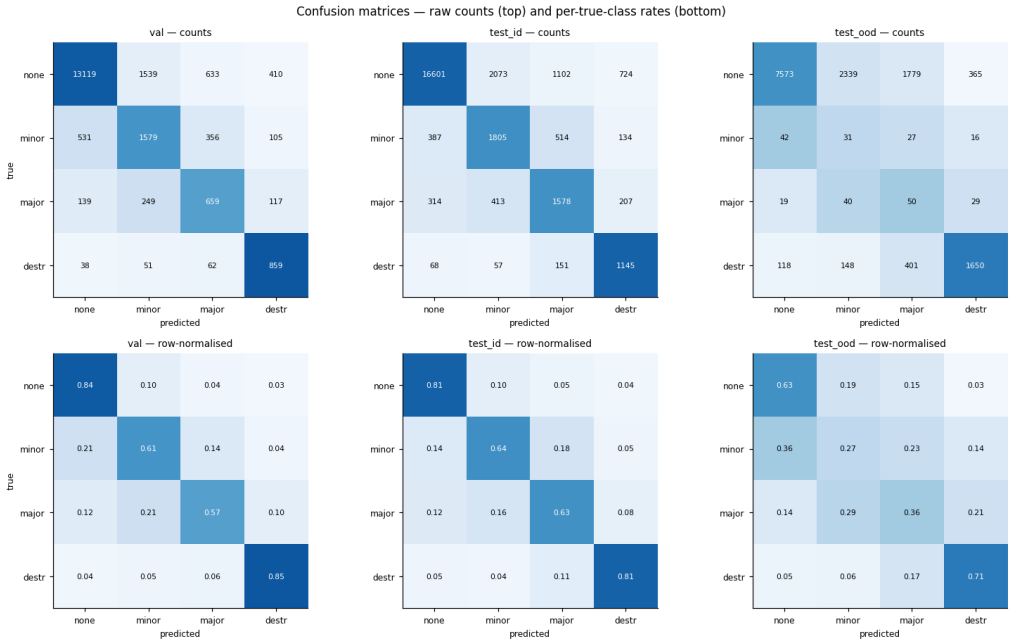


Figure 11: Confusion matrices, raw counts (top) and normalised per true class (bottom). The `test_ood` panel shows probability mass pushed off the diagonal in one direction — toward greater severity — rather than scattered.

it closes, that is the strongest result the project can report.

3. **Few-shot ablation**: add 1–5 % of wildfire scenes to training and re-measure the gap.
4. **Damage map** for a held-out event, using the per-building WGS84 coordinates already present in the labels.
5. **Part 3**: overlay predicted damage with socioeconomic data.

Reproducibility. The pipeline is a single notebook that reads the frozen split file, runs on Kaggle or locally, and writes all artefacts — patch arrays, cleaning ledger, occlusion candidates, training history and result tables — to disk. Seed 42 throughout.

Contributions. Artur Zavistovskyi: exploratory analysis, data cleaning and quality diagnostics, patch pipeline, pretrained model, evaluation, this report. Bhuvanesh Dinesh Wadhwani: dataset indexing and the train/val/test split design used here. Yitian Zhou: pretrained model (joint). Anastasiia Khitrova: from-scratch CNN and impact analysis (in progress).

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